

AGILE FROM FIRST PRINCIPLES

Lynda Girvan and Simon Girvan



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Agile exists as both a collection of practices and the principles that underlie those practices. Teams need both. But practices change over time—today's state of the art is passé tomorrow. And while Agile practices may change, the principles never do. That's what makes this book so valuable. Lynda and Simon provide plenty of time-proven practical advice—specific things you can begin doing immediately. But practices are always presented with a helpful dose of principles. You'll finish this book knowing what to do today and how to adapt those practices in the future.

Mike Cohn, *Mountain Goat Software, Co-Founder of the Agile Alliance and the Scrum Alliance*

First principles are regularly overlooked when people start with Agile, and also seemingly forgotten by many who profess to be experienced with Agile. This book delivers both a fantastic introduction and very insightful reminders as to why these first principles are essential to 'being Agile', all supported with real experience and insightful research. Essential reading.

Julian Holmes, *Principal Transformation Consultant, Thoughtworks*

Agile from First Principles is an essential compendium for delivery professionals, intuitively weaving together concepts from Agile, Lean and product development disciplines. It provides accessible expertise for teams starting their journey, insights into the daily challenges of working in rapidly changing environments, the spectrum of practices available to experienced Scrum Masters, and a library of nuanced learnings, grounded in the fundamentals of modern product delivery that will resonate with leaders. A highly recommended addition to an Agile coach's toolkit.

Tom Hoyland (*Expert Coach*), *Principal Agile and DevOps Consultant, That Agile*

This book is essential reading for anyone practising in an Agile context. The authors share their vast experience and knowledge in a compelling and accessible way starting at first principles. Along the way they summarise key concepts, techniques and 'antipatterns' (warning signs). The book is written in a refreshingly hands-on, practical way which cuts through the buzzwords and gets straight to the point. Highly recommended!

Adrian Reed, *Business Analyst at Blackmetric Business Solutions, Author of Business Analyst (BCS)*

This is a convincing commentary on successful Agile methods. *Agile from First Principles* gets to the heart of what being Agile really means: a focus on people and practical solutions. It explains why the Agile Manifesto and the principles behind it matter so much. An excellent orientation for beginners which will help avoid common pitfalls. For experienced hands, a fundamental study of wide application, Agile leadership and culture.

Toby Jones, *founder of the Accelerated Capability Environment (the government's innovation engine)*

This is an excellent book for people both starting their Agile journey and those experienced practitioners who are looking to either deepen their knowledge or in need of a little reminder of the core principles that make Agile such a powerful and effective delivery approach. The book has a great blend of background context and practical examples, demonstrating why Agile is so effective in today's information and technology-based world.

James Burton, *Director Consulting Expert and UK Career Champion for Agile at CGI*

Agile from First Principles is the guide for advancing your Agile journey. The book, as its name implies, goes from principles to specific techniques in a framework-agnostic manner making it a valuable resource irrespective of whether you use Scrum, Kanban, XP or any of the scaling models. This book is highly recommended for both Agile beginners and experts.

David Bulkin, *Managing Director, Grow-Lean LLC*

Some books can't be written until most other books on the subject already have been written. *Agile from First Principles* is such a book. In an easy-to-understand way, the book presents the journey Agile has taken from its first stumbling steps to where it is today, having penetrated much of industry and academia. 'Now this is not the end', as Churchill might have said, but we are at a point in time where an elegant summary of the mess in which Agile is will help us to successfully move forward to a better future. And the book points to this future with discussions on leadership, the challenges of scaling and a subject closer to my own heart, Essence.

Dr Ivar Jacobson, *Chairman and CEO, Ivar Jacobson International*

Agile from First Principles is the perfect guide to understanding Agile and what is truly different about Agile approaches. The in-depth explanation of values and principles is far from theoretical. Lynda and Simon use loads of examples, as well as common traps and anti-patterns, to help you understand what it really means in practice, going far beyond the usual applications and methods (they do an excellent job covering them too, by the way). This book has already changed how I teach Agile!

Joakim Sundén, *Agile Transformation Coach/Specialist,
Co-creator of the 'Spotify Model', Co-author of Kanban In Action*

As the title promises, Lynda and Simon introduce the underlying principles and values of Agile, emphasising the benefits of adopting an Agile mindset and attitude, alongside a straightforward guide to popular Agile methods and tools used within software development. Anti-patterns identified throughout the book help challenge common attitudes and behaviours that can limit and sabotage our ability to be Agile. The book invites the reader to reflect on what we need to do more of, but also what we need to do less of, and stop doing to improve our agility! An easy-to-read guide for anyone interested in understanding the fundamentals of Agile and identifying ways to improve their own agility personally, within their team, and as a leader.

Belinda Waldock, *Being Agile, Author of Being Agile in Business*

An expertly crafted go-to manual for both seasoned professionals and those just starting out in the world of Agile. An easy read, full of practical hints and tips that will empower you to overcome the issues that hinder so many Agile teams. Well written and full of rich examples, it is sure to become your primary Agile reference book.

Martin Maya, Founder, Amatis Training Ltd

Finally, everything is in one place. *Agile from First Principles* has quickly become my go-to Agile resource book. I have long been a fan of Lynda and Simon's work and their endless comprehensive knowledge of Agile models, frameworks, values, principles, examples and thought leaders. As expected, this book does not disappoint. This book is well structured and pitched perfectly for anyone entering an Agile environment or those who are experienced and looking to refresh and expand their knowledge.

Kylie Yearsley, Director, Agile Games Ltd

As the Agile movement enters its third decade, it has grown and moved into dimensions and directions never thought of by the writers of the Agile Manifesto. The success of Agile as a business has often corrupted the original ideas behind Agile, imposing it in domains where its application is not the optimal solution to the problems at hand. The questions of where Agile practices and techniques make sense, where they are out of place, and where and how they might be altered and extended are valid questions that require reflection on the original ideas and intent of the Agile founders. This book, written by two knowledgeable and experienced Agilists, provides the foundation for self-reflection on the basic principles and the what and why of Agile, to help ensure that future advancements maintain both the word and the spirit of the Agile Manifesto.

Joseph Pelrine, Agile Psychologist

This timely book exposes many stubborn myths around Agile and highlights the potential dysfunctions when seeking to achieve greater agility. In jargon-free language, it outlines a path for leaders and practitioners alike to thrive in an ever more turbulent climate. I would highly recommend this book to anyone seeking to move beyond tools, and to understand the true spirit of the Agile movement.

Karim Harbott, Co-Founder, Agile Centre,
Author of *The 6 Enablers of Business Agility*

AGILE FROM FIRST PRINCIPLES

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Further information

BCS, The Chartered Institute for IT,
3 Newbridge Square,
Swindon, SN1 1BY, United Kingdom.
T +44 (0) 1793 417 417
(Monday to Friday, 09:00 to 17:00 UK time)
www.bcs.org/contact
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Lynda Girvan and Simon Girvan

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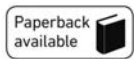
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ABOUT THE AUTHORS

Lynda Girvan is the head of Agile and Business Analysis for CMC Partnership Consultancy Ltd, which specialises in Agile and business transformation services. She is a Chartered Fellow of BCS, BCS subject matter expert in Agile and business analysis, holds the coaching certification Certified Scrum Professional with the Scrum Alliance and is working towards Certified Enterprise Coach. Lynda has extensive experience of adding value to organisations, from coaching Agile development teams to advising and coaching on board-level change programmes. Lynda's particular strength is applying her knowledge of change, business analysis and Agile to help organisations overcome challenging business problems, such as Agile transformation and organisational agility. Lynda co-authored the BCS book *Agile and Business Analysis*¹ and is an experienced and keynote speaker on these subjects at international conferences.

Simon Girvan is a principal consultant and Agile coach with Ivar Jacobson International, where he helps organisations to improve teams and products agnostic of the framework they are using. He holds the coaching certification Certified Scrum Professional with the Scrum Alliance and is working towards Certified Enterprise Coach in addition to being certified in various other methods, including Scrum, Kanban, Scrum@Scale, LeSS and SAFe. He is a Chartered Fellow of BCS and the Institution of Engineering and Technology (IET), and is a BCS subject matter expert in Agile. Simon has significant experience working with teams, leaders and organisations to help them adopt an Agile mindset. Having spent many years in the public sector leading Agile teams and coaching leaders, he is now applying his knowledge and experience in the private sector. Although he has an engineering background, Simon enjoys working with non-technical teams and with organisational agility initiatives. He is a regular speaker and author on Agile topics.

¹ <https://shop.bcs.org/store/221/detail/workgroup/?id=3-221-9781780173221>

FOREWORD

Over the last decade and more, my passion has been in supporting other Agile practitioners in their growth and development. I have spent much of that time helping those who work in service to their teams, organisations and, most importantly, colleagues in creating better workplaces – those more suitable for the challenges of the 21st century and more respectful of our humanity.

In that time, as a coach, consultant and trainer, I have collected many certifications, not for the recognition but for the platform they have given me to broaden my reach. My work as the managing partner of Agile Centre in both leading an organisation built on Agile principles and supporting other organisations in adopting those same principles comes with challenges many of you will have faced. It comes with great benefits too.

One of those benefits was meeting Simon and Lyn. What started as coaching and mentoring relationships quickly evolved into a relationship of peers. A meeting of minds and a differing of opinions. A place where I could challenge, and be challenged, across the broadest of horizons in the Agile discipline.

Over the last four years, I have had the privilege of witnessing both Lyn and Simon's knowledge, and understanding of the Agile mindset and philosophy crystallise. Already experts in other fields, it has been with great pleasure that I spent time working with them to synthesise their understanding and apply it to the context in which they have worked. This combination of clarity of thinking and deep experience and practice has come together in this book. The first on Agile, I suspect, of many to come.

I have enjoyed many conversations covering the esoteric to the mundane in the Agile world throughout the years. The ones that stay with me are not those where we spent time exploring some dark corner of thought related to team dynamics or social complexity. The ones that stick are invariably related to the core of Agile: the values and principles. The few words of the Agile Manifesto cut to the core of product development and knowledge working, and I find them still ringing true two decades later.

It will be no surprise then that the sections that stand out for me are the values and principles. Extraordinarily few books deal with the values and principles of the Agile Manifesto in a way that is both insightful and practical. Yet, they are the foundations on which we build our practice.

When introducing the Agile Manifesto, many texts give at best a second thought to the values, simply saying 'we prefer the ones on the left' or something equally as helpful. Instead, Simon and Lyn have tackled these powerful statements in a new and insightful

way. They examine each value, look deeper into the meaning, uncover why many organisations and teams find it hard to live up to them and call out the anti-patterns and problems they have observed in working with teams aspiring to the values.

Suppose you have found this book during your first steps on the Agile journey, congratulations. Adopting a new way of working, especially one as alien to the typical organisation as Agile is, is a daunting task. Take heart that practitioners such as Lyn and Simon have gone before you and drawn a map to help you on your journey. Your journey won't be the same, but there will be similarities. Please read the text carefully, make notes, take your time to understand the concepts and how they relate to your team and organisation. Which practices and ideas resonate with you? Which will be harder to bring to life?

If this book found you further on your Agile journey, you may look at the title and dismiss the content as irrelevant. I implore you to reconsider. As practitioners, we fall foul of hubris as easily as anyone else. If nothing else, revisit the values and principles and create time to reflect on what Simon and Lyn have to say and how your views have changed over the years of practice. What has changed for you? Why?

In the time I have known Lyn and Simon, we have debated every aspect of Agile conceivable, from the humble user story to the implications of true business agility on organisations and society via change management and business analysis. And I've enjoyed every minute.

In your hands is an invitation to join that discussion.

John McFadyen
Managing Partner, Agile Centre

ABBREVIATIONS

2FA	two factor authentication
3Cs	card, conversation, confirmation
API	application programming interface
BDD	behaviour driven development
CD	continuous deployment
CI	continuous integration
CTO	chief technical officer
DDaT	Digital, Data and Technology Framework (UK Government)
DoD	definition of done
DoR	definition of ready
DSDM	Dynamic System Development Method
GDPR	General Data Protection Regulation
HiPPO	highest paid person's opinion
HR	human resources
INVEST	independent, negotiable, valuable, estimable, small, testable
LeSS	Large-Scale Scrum
MI	management information
MMP	minimum marketable product
MoSCoW	Must have, Should have, Could have, Want to have but Won't have this time
MUST	Minimum Usable SubseT
MVB	minimum viable bureaucracy
MVP	minimum viable product
NFR	non-functional requirement
OKRs	objectives and key results
OMG®	Object Management Group®
PDCA	plan-do-check-act
PDSA	plan-do-study-act
PI	planning increment
RAD	rapid application development
RUP	Rational Unified Process

SAFe®	Scaled Agile Framework®
SCARF	status, certainty, autonomy, relatedness, fairness
TDD	test driven development
UI	user interface
UML	Unified Modelling Language
UX	user experience
VUCA	volatile, uncertain, complex and ambiguous
WiP	work in process (Lean), or work in progress (Kanban)
WSJF	weighted shortest job first
XP	extreme programming

GLOSSARY

Acceptance criteria: The criteria that specify whether a feature or work item will be accepted by the customer.

Agile board: A physical or electronic board that displays the current work of the team. A public way to share the work of a team.

Agile Manifesto: The result of the meeting that defined Agile and described it as a set of values and principles.

Agile practice: A concrete, specific thing that Agile teams do that helps them with an Agile mindset, includes techniques, approaches, processes, patterns, models or maps.

Anti-pattern: A pattern that describes a way something is done, organised or happens that is detrimental to the intended outcome.

Application programming interface: A set of functions or services that allow separate software systems to connect and interact with one another.

Backlog: A list of work items or ideas that could help achieve value for the customer.

Backlog refinement: Periodic review of the backlog to keep it up to date.

Behaviour driven development: A way to provide a shared understanding between the different perspectives of developer, end user and tester.

Big Bang delivery: Waiting until the entire product is complete before the customer can use it.

Big room planning: An event used in scaled approaches to bring together multiple teams for collaborative planning purposes.

Component team: A team that works on a single component as part of a wider solution, in contrast to a Feature team.

Context diagram: A diagram used to show the interactions between a product, or structure, and the things outside that product/structure.

Continuous deployment: Deploying new capability as soon as it is available, often combined with continuously integrating small pieces of work frequently.

Continuous integration: Integrating work into the main workflow as soon as it is complete. Often done with very small pieces of work, perhaps many times a day.

Customer journey map: Technique that maps the path a customer takes to achieve a goal or objective.

Daily stand-up: Daily team event to discuss and share what they have been doing, what they will do next and any blockers. Often conducted while standing to help ensure it remains short.

Definition of done: Criteria to verify that quality-driven activities are in place and completed for a work item.

Definition of ready: Criteria that describe what must be in place before work can commence.

Development team: The people in the team that are committed to creating any aspect of a usable increment in each iteration.

DevOps: A contraction of Development and Operations. The combination of cultural philosophies, practices and tools that bring together development and operations in the same team in order to deliver software at a faster pace.

Empathy map: Technique to help teams obtain a shared understanding and empathy for others and improve customer experience.

Empiricism: The theory that all knowledge is based on experience derived from the senses.

Essence: An open standard for defining methods and practices.

Feature: Functionality within a product that offers significant value to the customer.

Feature team: A team that delivers working features for the customer, in contrast to a component team.

Functional requirement: A way to express what the product needs to do.

Gantt chart: A popular way to describe a project plan where each activity is shown as a bar against a schedule. Tasks to be performed are on the vertical axis and time duration on the horizontal.

Increment: A step towards a usable product.

Iteration: A time-bound cycle, usually within a broader delivery life cycle, delivering a version of the product or service.

Kaizen: Japanese word for continuous improvement or 'change for the better'.

Kanban board: Physical or digital board used by teams to visualise the flow of work by the team to provide value to the customer.

Kanban Method: A method to optimise flow of work by a team.

Lean wastes: The seven wastes, identified by Taiichi Ohno, that should be identified and minimised in order to speed work by a team.

Linear approach: A software delivery approach that completes each step sequentially before starting the next (e.g. Waterfall approach).

Minimum marketable product: The minimum set of functionality for a product that could permit the product to be sold or used operationally by a customer.

Minimum viable product: The minimum set of functionality for a product such that the product can be used to run experiments and generate feedback.

Non-functional requirement: A way to express how well the product/programme will achieve functionality.

Persona: An archetype used to understand the motivation and drivers from a customer perspective.

Planning Poker®: An Agile estimation approach based on Wideband Delphi trademarked by Mountain Goat Software LLC and widely used by Agile teams.

Product: The solution that Agile teams are producing.

Product Owner: The role in Scrum that is accountable for maximising the value of the product resulting from the work of the team. Also widely used in non-Scrum teams.

Programme increment: An 8–12 week planning horizon used within SAFe.

Refactoring: The practice of restructuring existing solutions (e.g. computer code).

Requirement: Feature or need that has been requested by a stakeholder.

Retrospective: Regular event enabling the team to reflect, over a specified time period, on their ways of working and identify opportunities for improvement as part of their continuous improvement.

Roadmap: A way to describe the vision and/or goal for the work and a rough order.

Scrum: The most commonly used Agile framework for development teams.

Scrum Master: The role in Scrum that is accountable for establishing Scrum as defined in the Scrum Guide.

Shu Ha Ri: A Japanese concept that describes three stages on the path to mastery of a skill.

Spike: A short, time-bound task within an iteration that is not critical to the iteration goal. Often used to restrict time spent on open-ended tasks like research.

Stakeholder: Anyone who has a vested interest in, or is affected by, the product.

Story Points: A unit of measurement used by a team to size user stories in a Product Backlog to aid planning and forecasting.

The three pillars: The 'Empirical Scrum Pillars' from the Scrum Guide: transparency, inspection, adaptation.

Time boxing: The act of restricting a task, activity or event by a time limit.

T-shaped professional: People who are competent in a range of skills and have one or two areas where they have deep expertise.

Two factor authentication: A security feature that requires a user to have an additional form of identification to log into a system beyond just a username and password. For example, a token, mobile device, authentication app, and so on.

Use case: A lightweight, iterative way to describe a specific situation in which a product or solution could potentially be used.

Use case diagram: Model to visualise and express the scope of the endeavour showing use cases and their actors.

User role: Defines a single user group that interact with a product or solution.

User story: Short description of user value used in Agile approaches.

User story mapping: Technique to visualise user stories across the whole product or solution.

UX design: Design of the user experience elements of a product such as the graphical interface or web pages.

Velocity: The average number of Story Points (or other metric) per iteration that a team has completed.

Waterfall: A linear software development approach.

Wideband Delphi: A collaborative approach to estimate work.

Working solution: A working product that achieves value for the customer, but is not necessarily the final solution.

PREFACE

A quick search will reveal that there are already many books on Agile, covering a wide range of topics. Many focus on Agile methods and approaches; others on specific topics or techniques. Some books expect little prior knowledge; others expect at least an appreciation or understanding of the basic concepts.

In this book we have attempted to live up to the title *Agile From First Principles* and cater both for those new to Agile and more experienced practitioners who would benefit from a refresher. We incorporate both theoretical and practical application of Agile, combining our experience and stories with external references and guidance.

In this preface we provide a brief overview of Agile, provide indications about who this book is for, introduce you to our case study and provide a short synopsis of what to expect of the rest of the book.

WHAT IS AGILE?

Before we start, it is worth considering what 'Agile' actually is. At its simplest, it is an approach to developing products, often but not always software products. It applies an iterative and incremental approach, favouring the release of versions of the product throughout the development process, seeking feedback on them and incorporating that feedback into the development process.

This means that the development cannot be planned far in advance, as we can't second guess the feedback. Therefore, Agile teams must be able to plan for short horizons and be able to respond to change. Agile teams typically work in short cycles, a few days to a few weeks, before pausing to reflect on how they are doing and replan for the next cycle.

Agile teams use certain frameworks and practices to help them plan and deliver the product. They form teams and assign roles in a particular way, adopt high levels of collaboration and tend to operate relatively autonomously.

Already, we can see this is different from how many products are developed. We are used to projects deciding the product specification at the start and releasing the product, when it is complete, at the end.

Agile emerged in the early 2000s and initially had quite a niche following. However, over the intervening years it has grown in popularity and influence. As we write in 2022 it is

extremely popular, well known and widespread – not only for software products, but for solutions and projects of all shapes, sizes and types.

IF IT'S SO POPULAR, WHY DO WE NEED THIS BOOK?

You may be wondering why, if Agile is so widespread, this book is necessary? As you will discover, despite Agile being simple to understand, in practice it is very hard to do well.

Over the years, we have witnessed many teams struggle to apply Agile concepts consistently and successfully even with experienced teams. One of the reasons for this is that many people focus on the mechanics of Agile methods and don't give enough consideration to the underlying principles and values. The problem is that it's easy to apply the mechanical elements in ways that don't embody those values, and can even undermine them.

That's why we have called this book *Agile From First Principles* – we believe successful Agile adoption starts with a solid understanding of the fundamentals, and only once these are mastered can you expect to apply any Agile method or framework successfully.

WHO IS THIS BOOK FOR?

The short answer is that this book is for anybody who wants to understand Agile better. That includes those who are brand new to the subject. It also includes coaches like us who value a single book that reminds us of what is important and is a handy reference to a broad range of Agile concepts. This is the sort of book we would buy and use ourselves.

More specifically, there are certain groups of people to whom we particularly recommend this book.

Those new to Agile

This book provides a comprehensive introduction to Agile with a base of knowledge that makes learning and applying any Agile approach easier and more likely to succeed. The important elements are highlighted and reinforced together with examples and stories that help bring it to life.

Leaders and managers

Even if you are not involved in the day-to-day work of Agile teams, leaders and managers have an important role to play. Bringing out the best of Agile teams requires different sorts of leadership from what you may be used to. This type of leadership will feel uncomfortable and perhaps counter-intuitive. This book will help you to understand why it is important to lead differently and help you work out how. Of course, it will also help you to understand what's going on in Agile teams, demystify some of their language and behaviour, and help you to lead and manage them better.

Business stakeholders

Agile teams exist to provide value to customers. In order to provide the maximum value, those customers and other stakeholders need to understand how Agile teams work. Whether you are representing a direct user, or are a conduit to external customers and users, this book will help you to understand why Agile teams can be more effective and what you need to do to help them. This will usually involve some changes in behaviour and perspective. This book will help you with ideas and tools that can help you to achieve that.

Project managers

Being thrust into Agile can be confusing and bewildering for project managers, particularly those who are experienced and skilled. The Agile ceremonies, language and assumptions can be very alien when compared to best practice in project and programme management.

This book will help you to understand how Agile teams work and why they do some of the things they do. It will help you to understand how project managers can help Agile teams succeed, and why some things that project managers can expect may not be helpful for Agile teams. If you are taking on an Agile role, this book will help you to understand what is expected of you and give you the knowledge and understanding to perform it well.

Students and trainers of the BCS Foundation Certificate in Agile

The BCS Foundation Certificate in Agile is an excellent certification that gives a broad grounding in Agile concepts and practices without focusing on a particular method or practice. This book covers the whole syllabus, although it isn't written in quite the same order.

The 2020 version of the syllabus was written by one of this book's authors and one of its reviewers, so the perspective of the book represents what the syllabus intends to convey.

Agile practitioners and coaches

It may sound odd that a book that addresses first principles, and is suitable for novices, should also be valuable to experienced practitioners. However, there are two good reasons why this is the case and why this is exactly the sort of book we would buy for ourselves.

First, this book covers a lot of ground to explain the principles and practices of Agile. There are dozens of techniques, ideas, methods and practices described; many in brief detail with references to more detailed descriptions. As such, this one volume contains just enough information to be useful but references a huge body of knowledge that is invaluable to any Agile practitioner.

Secondly, even experts in Agile practices are not immune to falling into the traps and anti-patterns that we discuss. Having a reminder of the core principles of Agile to

hand helps to keep us grounded. The discussions on the values and principles will be particularly useful to help Agile teams improve.

WHAT TO EXPECT?

This book is a mix of fact, opinion, guidance and stories. Agile emerged as a short website in 2001 with just 68 words describing its values and a further 186 covering the principles.

We help you to understand those values and principles from our experience as Agile practitioners and coaches with many years of experience. We have tested our opinions with reviewers, colleagues, peers and others. We think that understanding what's behind Agile and what it means in practice is important. That's why we have devoted so much space and effort to explaining what we think they mean. We have illustrated this with examples of anti-patterns and problems we have experienced and use them to show not only why it's so easy to get Agile wrong, but ways that you can avoid making those same mistakes yourself.

We wanted reading this book to feel like a conversation with an experienced Agile coach. One who can explain new things to you, share their experiences, point you towards some good books or references, suggest new ideas and, most importantly, challenge you and make you think. You may not agree with all of our opinions, and you may speak to other Agile coaches who have different ideas. That's fine. Just treat this as new information to add to your existing knowledge and opinions and perhaps you will formulate a slightly new and evolved opinion. As you will read, we call this inspection and adaptation, and iterative development. It can apply just as much to ideas and beliefs as it can to products or processes.

It is worth mentioning that this book is unashamedly about Agile – the word is used over 900 times. This book is definitely about Agile delivery, Agile teams and for people interested in Agile. However, we are not trying to tell you that Agile must be the approach to everything, or that it will be the best choice for every endeavour. Agile is an excellent choice for many types of project and deliveries. It is the best choice for some; one of several good choices for others; and a poor choice for some. There are many situations where a well-executed alternative approach will be better than a poorly executed Agile approach. There are many scenarios where either a traditional or Agile approach can work well, provided they are executed well. Poor application of any approach will lead to poor results, even if that choice of method is ideal for the situation.

This book isn't about encouraging you to use Agile when it isn't appropriate. But any time you **do** choose Agile, this book will help you to execute it well and maximise the value you will achieve. By understanding the principles of Agile, you will also be able to judge when Agile will be appropriate to use and when it is not.

HOW TO READ THIS BOOK

We have written the book to flow logically: it starts with the background and history of the Agile movement, describes the main concepts, then goes into more detail on some

specific, important areas in the later chapters. Thus, you can start at the beginning and the book will make sense.

However, you can also jump into a chapter that interests you and it should still make sense without having read the previous chapters. For example, much of the leadership chapter stands alone, as does the chapter on managing the product. This does mean there is a little repetition, particularly in the chapter on Agile practices. This is because we feel it is useful to have a summary or introduction to a practice even when we later expand on it in more detail.

Later in this preface, we provide a summary of each chapter to help you work out where to start. The table of contents and index will also help. However, we would urge you to spend some time on the values and principles chapters. Even if you think you know the Agile Manifesto – and, perhaps, **especially** if you think you know the Agile Manifesto – these chapters contain a wealth of experience and discussion on how the values and principles can be applied well and, importantly, how they are commonly misapplied. We are confident even experienced Agile practitioners will find something of value.

AN INTRODUCTION TO AG STORES

Having an example always makes things easier to understand. In this book, we have tied most of the examples to a fictional product that we hope you will be able to relate to.

AG Stores is a chain of large superstores that sell a variety of products, including food, hardware and home decoration products. Having relied for years on a loyal customer base, AG Stores are experiencing increased competition from other shops and also from online competitors. They believe that introducing a loyalty programme will allow them to reward their existing customers, dissuade the customers from shopping elsewhere and provide a rich set of data that they can use to improve marketing, store layout and stock control.

To cater for some customers, they will need physical cards and telephone-based account management, but they also want an electronic solution that can be more tightly integrated to their other IT programmes, allowing real time data analytics and personalised content such as offers or adverts. The proposed solution will include:

- a mobile app for customers to manage their account, view offers and discounts and present at the till;
- a website for managing accounts and viewing discounts and offers;
- integration between the loyalty programme and the till structure so that points can be accrued and spent;
- data analysis components that can generate insights into customer spending habits, steer marketing campaigns, assist with financial analysis and help with product placement;
- offline support for customers who don't want to use the app, including physical membership cards.

This scenario allows us to explore many aspects of product delivery, including how non-IT elements (such as marketing) can be included. There are many ways to deliver such a solution, and many ways to split up the work. We don't pretend our examples are the perfect answer (if such a thing even exists) and there are many alternative ways we could have chosen. We hope that the examples will help bring some of the concepts to life.

A BRIEF WALKTHROUGH

In this section we provide a brief walkthrough of the chapters in this book to help you decide where to start.

Chapter 1 – The importance of Agile today

This chapter explains why Agile approaches are still important today – despite emerging over 20 years ago, the conditions that led to Agile are not only still present but have intensified. Agile has matured beyond its origins in software to many other fields, and modern leadership theories are converging with the types of leadership advocated by Agile approaches. Coupled with Agile's growing dominance in the technology sector, it is clear that Agile is still relevant and the ability to apply Agile well has never been more important.

Chapter 2 – The origins of Agile

Here we discuss the theories and practices that have influenced Agile, some of which date back centuries. We discuss how non-Agile teams plan and manage their work and tell the story of how the Agile Manifesto came to be created in a ski lodge in Utah in 2001.

Chapter 3 – Projects and products in Agile

One of the ways that Agile bucks the trends set by traditional development approaches is that it has a product focus rather than a project focus. In fact, not only is the term 'project' not even mentioned in many Agile approaches, we suggest its use can actually be harmful – encouraging an unhelpful focus on tasks and milestones and minimising the importance of responding to change. We explain how Agile delivery is incremental and iterative and this is how we can deliver more value. The 'Product Backlog' is introduced as an important artefact.

Chapter 4 – Understanding the Manifesto for Agile

This chapter, along with [Chapter 5](#), is one of the two cornerstone chapters in the book. The Agile Manifesto defines what we mean by 'Agile' so, to apply Agile well, it is critical that we understand what is meant by the Agile values to understand how we can apply and reinforce them.

A recurring theme throughout the book is that Agile is difficult to do well and we have seen many teams struggle. Even though the values look like common sense, they are not

always commonly applied. Therefore, we look at common challenges and anti-patterns we have seen through the lens of each value, showing how they can be reinforced or undermined. There are plenty of references and further reading offered here too.

Chapter 5 – Understanding the Agile principles

Similar to [Chapter 4](#), here we explain and examine each of the Agile principles in turn. We share our perspectives on what they mean and why each is important. We focus on particular words and phrases and drill down to consider them more deeply. This allows a far greater insight into what Agile means and how we can apply it ourselves and with our teams.

Additionally, we unpick the intent behind each of the 12 Agile principles and explore anti-patterns that may prevent you from applying Agile as it was intended.

Chapter 6 – Fundamental concepts in Agile

There are several common themes and concepts that run through the values, principles, core behaviours and mindset that underpin any Agile approach. This chapter introduces and discusses them. They are: empiricism; the three pillars of transparency, inspection and adaptation; lean approaches and wastes; incremental and iterative delivery; customer centricity; and self-organising, empowered teams. The chapter explains with some examples how these reinforce the values and principles.

Chapter 7 – Agile delivery

Agile teams need some defined approach to help them apply the values, principles and fundamental concepts to deliver value to their customers. There are several common patterns and techniques that are present in most Agile methods and frameworks. This chapter shows the delivery approach in a generic way, agnostic of any specific approach.

We discuss the overall life cycle; how to get started; how Agile delivery differs from a traditional approach; how to measure progress; getting to the first delivery of value; how to include continuous improvement; how iteration works; and how the team behaves and is organised. We introduce some of the other roles commonly found, including some that are specific to a particular method or framework.

Chapter 8 – Agile practices

Here we discuss and briefly describe a range of practices, techniques, methods, approaches and other ways of working that Agile teams commonly use. They are grouped into categories covering the following areas: leadership; ways of working; requirements, estimation; prioritisation; software development; and measuring success. We briefly describe two of the most common Agile approaches used by teams: Scrum and the Kanban Method.

Chapter 9 – Agile leadership

The easiest way to make an Agile team fail is to surround them with poor leadership and management that don't understand Agile. This chapter will help to address that by

helping leaders and teams develop the kind of leadership that will help Agile teams to deliver the best results.

We describe the evolution of leadership and how many management techniques that evolved in early 20th-century manufacturing are still prevalent today, despite the conditions that Agile teams operate in today being vastly different from those factories or mines. Agile leadership needs to be different; ironically, we look to a German field marshall from the 1800s for a description of what it should be.

For many leaders, effectively leading Agile teams requires a shift in mindset, and we identify and discuss our six key leadership outcomes that leaders should strive for: servant leadership; psychological safety; intrinsic motivation; trusted, empowered people; self-organising teams; and creating strategic alignment. Many of these are not unique to Agile leadership, and we share stories and references that show how these leadership outcomes bring value to many different types of teams. We talk about the importance of creating the right culture and why Agile coaching is a powerful path to success.

Chapter 10 – Managing the product

One of the biggest differences between traditional and Agile delivery is how the product requirements are understood and managed, in particular how the customer is involved and how teams cope with changes to the customer needs. This chapter looks in detail at how Agile teams understand their customers and manage their product and their stakeholders using techniques such as roadmaps, backlogs and story mapping. Some of these techniques have entire books to describe them, so there are lots of examples and references to help bring the ideas to life and point you towards further information.

Chapter 11 – Beyond the basics

Finally, in this chapter we introduce scaling. We have consciously left it until last because we think it is important to properly understand the basics before trying to scale.

It is also important to only introduce scaling when you need it, so we devote the bulk of the scaling discussion to ways that you can avoid the need to scale, even when you have large products or lots of teams. We show ways you can reduce the drivers for scaling by reducing complexity, decoupling everything, managing the product differently, splitting goals, refactoring, using feature teams and shortening the cycle time. If you really, really want to scale, we also cover that briefly.

A second practice that you should only embark upon when you understand the basics is tailoring a process. An aspect of tailoring is the use of specialist roles or teams that can sometimes be sensible to use.

We finish by describing in brief some of the more commonly used Agile methods and frameworks, including references to their source to allow you to find the current versions.

LET'S GET STARTED

We set out to write this book with the expectation that it would mostly involve writing down what we had learned and experienced through our years working with organisations, teams and individuals. In fact, it's been much more than that. Thinking through the Manifesto values and principles in detail, and spending time reflecting on ways that we have seen them reinforced and undermined, has improved our own understanding of them immensely. Similarly, researching more in-depth leadership and managing the product chapters has improved our understanding and knowledge of those important areas. As a result, this book is far more than an expression of our experience – it is a book that we will refer to, recommend and use in our day-to-day work with Agile teams and others.

We hope you will find it as useful and as illuminating as we have.

1 THE IMPORTANCE OF AGILE TODAY

It has never been more important for teams and leaders to understand Agile and how to apply it well

This chapter explains why Agile approaches are not only still important today but are more important than ever. We will explore why:

- Despite emerging over 20 years ago, the conditions that brought about the movement towards Agile development are still present and have intensified.
- Agile approaches are being applied in a growing range of areas and becoming commonplace, although they are also hard to do well and often applied poorly.
- Agile approaches, which began with software projects, are being applied to a wider range of business problems and are well matched to them.
- The secret to applying Agile well is to focus on acquiring a deep understanding of the underpinning fundamentals of Agile rather than just learning a framework.

WHY ARE WE HERE?

Before we start to consider the importance of Agile today, let's look back at why Agile emerged.

In the late 1990s, development of new IT systems was characterised by large, expensive projects that seemingly took a long time to deliver. Teams would often attempt to understand the totality of the solution before any of the ideas had been tested. The result of this was lots of time spent ensuring a full set of requirements were documented up front before a line of code was written.

This approach is reasonable when there is little change and when you can be confident that your requirements are right. This may have been true for some of the systems and projects in the late 1990s, such as tax systems or banking systems that had a lifetime measured in years, but even then it was still a bit of a gamble. It is much less true now. Today, even a tax system requires regular changes to keep pace with the changing world that it supports.

In the mid-1990s huge organisations such as Lockheed, Chrysler, Xerox, Honda and Borland were trying things differently and getting better results. As we will explore in [Chapter 2](#), those pioneers and innovators joined forces with others, also doing things differently, to create the paradigm shift of Agile development.

At the heart of this paradigm shift was the belief that reducing lead times, embracing changing requirements and improving customer buy-in was possible and would provide the ability for technology to create solutions faster than had been possible in the past. These core factors are not only still true today but have become orders of magnitude more significant.

We live in an increasingly VUCA world

The term VUCA emerged from the US War College in the wake of the collapse of the USSR and the end of the cold war. It was inspired by the leadership theories in the 1985 book *Leaders: Strategies for Taking Charge* by Warren G. Bennis and Burt Nanus.¹ They recognised that what had been a relatively stable and predictable state was suddenly anything but that; things had become volatile, uncertain, complex and ambiguous. This predicated an entirely different military approach.

What is meant by a VUCA world? It's one that has the characteristics set out in [Table 1.1](#).

Table 1.1 VUCA

Volatility	The world is constantly changing. Change, both large and small, is more unpredictable and happening more frequently. This instability results in unforeseen, dramatic changes with cause and effect difficult or impossible to analyse at present.
Uncertainty	It is more difficult than ever to anticipate events. Attempts to use historical data to predict what will happen next fail as the past is no longer a good predictor of the future. This makes it extremely difficult to plan with any certainty.
Complexity	The world is more complex than ever. Problems are layered, nuanced, have more dimensions and are harder or impossible to truly comprehend. Information to support decisions is incomplete, contradictory and untrustworthy. There isn't always a single right decision and, if there is, the answer is difficult to reach through analysis.
Ambiguity	There is rarely a single right answer or unique best practice; everything is shades of grey. Decisions, plans and advice exist in an environment of contradictory, missing and extraneous data. Operating like this requires courage when taking decisions, and a willingness to accept you may turn out to be wrong.

The COVID-19 pandemic that began in 2020 is the perfect example of a VUCA situation. The situation was complex and constantly changing; large amounts of data were available, but they were contradictory and incomplete. The political challenge of balancing economic impact, protecting society, preserving medical capacity and managing infection rates was an impossible task. Any solution that helped in one dimension caused harm in another. As more data emerged, previous decisions needed to be revised.

¹ Bennis, W. G. and Nanus, B. (1985) *Leaders: The Strategies for Taking Charge*. Harper & Row, New York.

It is easy to find other examples of a VUCA world, particularly with knowledge work (i.e. work that is non-routine and requires divergent and convergent thinking to solve problems). We have learned to construct complicated physical solutions such as bridges and electronics dependably, yet even the most experienced and skilled teams struggle to deliver IT or business change projects without exception reports, changes to requirements and rework. This isn't a failure of their management approach or poor quality implementations – this is because the problems being solved exhibit VUCA.

We find it hard to predict the future at the best of times. That's why IT projects are so difficult. They generally depend on user requirements being an accurate prediction of what users will need – yet, people are very bad at accurately predicting this. The problem is exacerbated when the data that are informing the prediction keep changing. This is true of non-IT projects, such as business change or service delivery projects, too.

When a problem is a VUCA problem, you cannot follow the current plan or analyse your way to a solution using traditional methods. When problems happen – the requirements change or the solutions don't work – it isn't generally a failure in planning or implementation; instead, it is often an acknowledgement that the problem wasn't capable of being analysed in advance. More or better analysis probably wouldn't have resulted in a successful delivery – it would likely have resulted in different things going wrong.

When we are confronted with VUCA problems, we need to solve them in different ways, and that's where Agile comes in.

Converging leadership approaches

The problems of a VUCA world are not restricted to areas where Agile is traditionally used, so it should be no surprise that the direction of travel of Agile leadership and more general management theories are converging. Leadership experts and gurus who have never even heard of the Agile Manifesto are coming to conclusions about how best to lead teams that are very similar to those of Agile thought leaders.

Joost Minnaar and Pim De Morree quit their corporate jobs and embarked on a mission to meet the people behind the world's most inspiring organisations and understand what makes them think differently. They documented their findings on their website² and in their inspiring book, *Corporate Rebels*.³

The organisations that inspire them include clothing companies, bus manufacturers, public sector organisations and banks. Few would describe themselves as an 'Agile organisation', yet the practices, behaviours and cultures they have created resonate strongly with those described in the Agile Manifesto.

² <https://corporate-rebels.com>.

³ Minnaar, J. and de Morree, P. (2020) *Corporate Rebels: Make Work More Fun*. Corporate Rebels, The Netherlands. <https://corporate-rebels.com>.

David Marquet's book, *Turn The Ship Around*,⁴ describes how he transformed his leadership from one of command and control into what he calls 'Intent Based Leadership'. Team members at all levels are empowered to take decisions; rather than waiting to be told what to do, they know what the overall vision is, decide what they need to do to move towards it and share what they intend to do with their colleagues and leaders. What's even more remarkable is that Marquet did this in the most surprising of situations – as commander of a nuclear-powered submarine in the US Navy. Although not labelled as an 'Agile' transformation, many of the changes Marquet made are the same ones we would recommend to organisations trying to be more Agile.

Agile leadership isn't just for small teams either. Jeff Sutherland, one of the creators of the most widely used Agile approach, Scrum, was part of the team that applied Agile approaches to the development of a military jet fighter – the JAS 39E Saab Gripen.⁵ As on David Marquet's submarine, decisions were given to the team to make wherever possible. They operated to short three-week iterations with higher level coordination happening quarterly – orders of magnitude more frequent than most military procurements. They had a strong focus on continuous improvement and ensured frequent customer engagement by having pilots available on the same site to provide feedback.

In his book *Implementing Beyond Budgeting* and his work with the Beyond Budgeting Institute,⁶ Bjarte Bogsnes describes how it is possible to run huge traditional businesses in radically different ways. He explains that organisations don't need to focus on short-term financial targets or plan around fixed dates (such as financial year boundaries). They don't need to set targets and cascade them to teams or control their people with detailed rules and performance evaluation. Instead, they can have teams with high degrees of autonomy over their work that organise around business-centric rhythms and events. They draw down resources when they are needed and are aligned and driven by high-level strategic objectives. Organisations operating successfully like this include large banks, petrochemical companies and manufacturers.

When author Dan Pink researched 50 years of behavioural science and psychology to understand what motivates us, he discovered that it's not what many people think. His book⁷ explains why trying to motivate teams with money or promotions not only doesn't work, it can actually make us perform more poorly, especially for knowledge work. Instead, he found the key drivers of intrinsic motivation are autonomy, mastery and purpose.

4 Marquet, L. D. (2013) *Turn the Ship Around*. Penguin, London and New York.

5 Furuholm, J., Segertoft, J., Justice, J. and Sutherland, J. (2015) Owing the sky with Agile: Building a jet fighter faster, cheaper, better with Scrum. [scruminc.com. https://www.scruminc.com/wp-content/uploads/2015/09/Release-version_Owing-the-Sky-with-Agile.pdf](https://www.scruminc.com/wp-content/uploads/2015/09/Release-version_Owing-the-Sky-with-Agile.pdf).

6 Bogsnes, B. (2016) *Implementing Beyond Budgeting*. John Wiley & Sons Inc, Hoboken, NJ. <https://bbbt.org/>.

7 Pink, D. H. (2011) *Drive: The Surprising Truth About What Motivates Us*. Canongate Books Ltd, Edinburgh.

These are just a few examples of advances in leadership and management theory that share many attributes with an Agile mindset. We will explore them in more detail in [Chapter 9](#). Just as these approaches can support and enable Agile teams, so too can Agile approaches be valuable to leaders in any sector. As these examples show, you don't have to be applying an Agile methodology to benefit from Agile leadership approaches.

Technology – friend or foe?

Technology is advancing at an incredible pace. In particular, the development of cloud technologies, and their ability to rapidly provide processing capacity, have revolutionised our ability to deploy IT solutions. This advancement in technology has tremendous benefits, but it also brings threats. User expectations are shaped by their experiences with other products. We interact with high quality, easy to use products in our social lives and we expect the same at work.

The barrier to entry for competitors is set by their rivals; and the ability of new entrants into a market to change, evolve and respond to new opportunities can give them a competitive advantage over existing incumbents. We have seen examples of this behaviour in banking, with the arrival of digital-only challenger banks; in media, with streaming services and digital journalism; and in the public sector, with the digitisation of formerly paper-only forms. Starling Bank went from hiring its first technical team to launching current accounts in less than a year.⁸ That's a pace that traditional banks could only have dreamed about.

This advance in the pace of technology makes it hard to keep up – particularly if you are working on systems that take months or years to get to market; any technology choice you make will almost certainly be out of date before it gets used. Agile's focus on releasing value quickly and iterating frequently is well suited to this trend in technology.

Another attribute of Agile development is the expectation that designs and architectures evolve and are refactored. Iterative development means that a given component or function is frequently revisited to add new capability. Each time this happens is an opportunity to refactor the component to take advantage of advances in technology.

Agile products constantly evolve.

THE DOMINANCE OF AGILE

In recent years, Agile approaches have become dominant for IT. In 2007, software vendor VersionOne began an annual 'State of Agile' survey. The first version had 722 respondents, of whom 84 per cent were in organisations that had adopted Agile in their software development. Their average length of experience with Agile was less than two years. In the 15th survey, published in 2021, 95 per cent of organisations were practising Agile, with around a third doing so for more than 5 years.⁹ A quarter of those surveyed

⁸ <https://www.starlingbank.com/about/road-to-starling/>.

⁹ Digital.ai (2021) 15th annual state of Agile report. <https://stateofagile.com/>.

were in organisations with over 20,000 staff, showing that the move towards Agile over traditional approaches is not just for small startups.

But it is not just the development of software products where Agile is growing. A quick search for books published on Agile will show that Agile approaches are being applied to business analysis, systems engineering, human resources, IT service management, business change, sales, marketing, organisational culture and, of course, project management. Applying Agile to project management means that pretty much anything that could be a project can now be 'Agile'.

One area of growth is in applying Agile approaches at an organisational or business level, operating whole teams, divisions or companies along Agile principles. This is usually because organisations realise that they are operating in a VUCA world and it isn't just their products and services that need to respond to change and complexity, but the organisation itself.

Agile organisations don't just optimise their current state, they adapt, pivot and revolutionise their whole mission. Well publicised examples include Netflix, FujiFilm and Amazon, but the global COVID-19 pandemic provided an opportunity for Agile organisations to change and flourish in ways that just aren't possible when you are driven by three- to five-year strategic plans and fixed approaches.

Some businesses that rely on face-to-face interaction with customers decided to sit out the pandemic and wait for the world to get back to normal. Others didn't. British farmhouse cheese specialists The Courtyard Dairy¹⁰ saw demand for cheese tastings plummet. They reacted by changing to an online model where they post the cheese and wine out to participants who join their experts for an online tutored cheese tasting.

Many training organisations pivoted to online training. Some just moved their existing courses online; others, including Agile Centre,¹¹ completely revised them to adapt to the differences in learning online, which had both advantages and disadvantages. For instance, they found that learning is more effective with two half-day sessions rather than one full day.

This ability to pivot demonstrates an underlying belief in the principles set out by the Agile Manifesto and provides the flexibility within the organisation to make such changes; changes that other organisations, big and small, have found difficult to make.

SIMPLE TO UNDERSTAND, DIFFICULT TO MASTER

If Agile is being applied to everything, does that mean that it is successful everywhere? Sadly, it doesn't. Agile projects also struggle and fail, just like any other type of project. However, that does not necessarily mean that Agile is not suited to those situations. More often we see examples where Agile is a good choice, but poor or naive implementation

¹⁰ <https://www.thecourtyarddairy.co.uk>.

¹¹ <https://www.agilecentre.com>.

has caused problems. This is most common with inexperienced teams, but we frequently see badly executed Agile approaches by experienced and well-trained people.

In particular, it is common to see Agile teams revert back to traditional linear project approaches even though they are following an Agile approach such as Scrum. Such projects may default to capturing all the requirements at the start, then spend months of development effort without producing anything that a customer can trial or use. These projects inevitably end up with delays as the requirements and understanding evolves, resulting in changes and additional work at the end to get approval for a formal 'go live'. All too often the customer buy-in so readily available at the beginning has been lost, and trust between the customer and the development community is broken. This demonstrates that blindly following an Agile approach is not the answer. Rather, the fundamental principles and behaviours need to be embraced by people in the project and in the work that is carried out.

This paradox – Agile is simple to understand yet difficult to apply – runs through many of the challenges teams and organisations face when trying to adopt Agile frameworks or methods and is a theme we will revisit.

Why is it so hard?

There are many reasons why Agile is so difficult to adopt. It suffers from the same factors that any large change suffers from: organisational resistance, Conway's Law¹² and other systemic issues. Yet, with Agile's people and team-centric focus, you may be surprised that, even within a team, Agile adoption is still difficult. We have identified three main factors within teams that contribute to Agile failures.

One – Comfort in the familiar

The Agile Manifesto is only 68 words and seems like common sense. This can lull us into a false sense of security. If it's common sense to us, then surely we will be adhering to its values by default? An Agile mindset, however, is quite a large paradigm shift from where most people's default behaviours lie. We are comforted by following plans and processes; we are used to there being a single correct answer that we can find if we try hard enough; we have been consistently rewarded for individual performance over team performance throughout our academic and work careers; we value big projects and big change more than small increments and partial solutions.

These are just some of the things that make us naturally resistant to the values and principles of Agile and why it is so easy for us to veer away from good Agile practice without realising it.

Two – Too quick to customise

Teams can often rush to change and customise an approach too quickly. This is partly the fault of some Agile trainers and coaches who (rightly) say that there may be some 'tailoring' required to match the approach to the situation. While this is true, most Agile

¹² Melvin Conway observed in 1968 that 'Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure.' Conway, M. E. (1968) How do committees invent? *Datamation*, 14 (5), 28–31.

practices are quite widely applicable and seldom need much 'tailoring' when they are applied correctly. The problem is that they are often not applied well. Therefore, before tailoring and changing (or omitting) an element of an Agile approach, we would always advise making absolutely certain you are applying it properly in the first place.

Three – It would be easier without the people

The third factor that makes Agile hard in practice is that it is a highly people-centric approach, and people are complex, unpredictable and wilful. Agile doesn't have formulae and detailed analysis that will tell you the right answer; it gets people together to share knowledge and understanding before selecting a way forward that might be right or might not. It doesn't put individuals in absolute charge – it expects teams to collaborate to reach decisions – and it doesn't assume things are going to go well – it expects problems to arise and for teams to identify constant improvements to the way they work. All these things are unusual for many people, and counter to much of our experience and training to date. That makes it hard to apply well.

These three reasons are why getting a firm grip on the fundamental principles of Agile is so important. When we deeply understand Agile and why it works so well when applied properly, we are in a far stronger position to identify when things start to go wrong and adapt. When we understand why Agile approaches work, it is easier for us to hold faith when it feels uncomfortable and to make adaptations that reinforce the Agile values rather than undermine them or revert to the familiar.

SUMMARY

The Agile Manifesto is coming of age. It is no longer only relevant to software products or small organisations. A much wider range of organisations, leaders, teams and projects can benefit from a more Agile approach.

Through its focus on dealing with complexity and change, there are more and more business problems that suit an Agile approach. Through its focus on people, interactions and collaboration it is converging with evolutions in leadership theory, making Agile relevant to business change and enabling change to organisational culture and behaviours. But it won't be easy. Adopting an Agile mindset will challenge decades of cultural norms and business behaviours. There is a strong tendency to 'snap-back' to traditional approaches, which are comforting and familiar. Even when we know they don't work well, it is easy to presume that this time might be different. They appeal to our logical brains. It makes sense that we should be able to conduct the right analysis, define the right plan and execute it to succeed; however, now, more than ever, we are realising that it's just not possible to do that. Even for relatively simple problems, there are too many variables, too little we can trust to remain the same and too many people involved. This environment is where Agile approaches thrive. For example:

- having a clear vision but achieving it with small iterations of value;
- continually testing whether you are doing the right thing through real-life problems;
- expecting to be wrong and having the ability to adapt and change;
- assuming you can get better and looking for ways to improve;

- trusting people to do the right thing and empowering them to make decisions;
- sharing what you are doing;
- being transparent and asking for feedback so you can get better;
- stopping when you aren't making things better, or where your time is better spent on a different problem.

There isn't anything software-specific about the above statements. They can apply in a wide range of situations and be applied at any level of an organisation. This book will delve into this in more detail and help you to understand Agile principles so that you can apply them to your situation, whatever it is.

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AGILE FROM FIRST PRINCIPLES

Lynda Girvan and Simon Girvan

Agile principles and practices transform the way organisations carry out business and respond to change. To realise success, an agile mindset needs to be adopted throughout an organisation, not just within the IT department.

This book is an introductory guide for individuals and teams, regardless of role, to applying Agile from first principles, encompassing agile values, mindset and practices. Practical examples are used throughout to illustrate agile theory with real-world context. This book is ideal for those wanting to achieve the BCS Foundation Certificate in Agile.

- Understand the importance of Agile today and the benefits it can bring if properly applied
- Explore the Agile principles and values, and the core concepts that emerge from these
- Discover Agile practices, enhancing your Agile mindset
- Learn how Agile leadership differs from traditional leadership and how to become an effective agile leader
- Focus on managing the product, the single most important artefact in Agile

ABOUT THE AUTHORS

Lynda Girvan is a business consultant, trainer, coach, author and practitioner. Her particular strength is applying Agile mindset and concepts outside software development to help organisations become adaptive.

Simon Girvan is an experienced Agile coach with over 20 years experience as a practitioner and Agile leader working mainly in the public sector.

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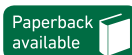
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